MATP: Memory-Augmented Tool Policy

A Unified Framework for Long-Horizon Autonomous Agents

Causal State Invalidation · Temporal Decay · Macro Distillation · Policy Learning

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The Long-Horizon Problem: Why Standard Agents Fail

1. The FIFO Context Trap

- Context pruning evicts early steps first.
- Foundational invariants (schemas, secrets) are lost.
- **Impact:** 75% critical early state forgotten over 40 steps.

2. The Naive RAG Trap

- Unversioned vector search indexes mutating files.
- Semantic similarity contains no concept of invalidation.
- **Impact:** 40% stale-state leakage (superseded schemas float up).

3. Procedural Thrashing & Schema Errors

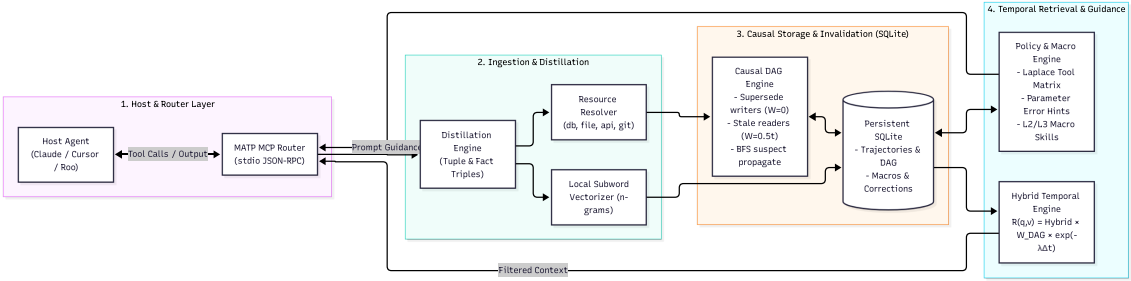
- Static system prompts lack dynamic empirical feedback.
- Agents get trapped in repetitive trial-and-error tool loops.
- Syntax errors (e.g. SQLite syntax quirks) repeat endlessly.

The MATP Monorepo Solution

A unified local MCP system combining:

- **Causal DAG** state tracking (W_{DAG}).
- **Temporal decay** & adaptive λ learning.
- **Macro distillation** & self-healing policies.

Unified System Architecture: End-to-End Pipeline



1. Ingest & Distillation

Distills raw tool outputs into structured tuples; extracts canonical resources. 13.5× compression.

2. Causal Storage (SQLite)

Tracks resource versions, supersedes older writers, marks readers stale, propagates taint via DAG.

3. Retrieval & Policy

Hybrid scoring with exponential decay $e^{-\lambda \Delta t}$, Laplace candidate matrix, auto-correction hints.

Contribution 1: Versioning with Piecewise Temporal Scoring

Piecewise Retrieval Formula $R(q, v_i)$

$$R(q, v_i) = \begin{cases} 0, & \text{if } v_i \text{ is superseded or depends on invalid state,} \\ S(q, v_i) \cdot W_{\text{DAG}}(v_i) \cdot e^{-\lambda(t_{\text{curr}} - t_i)}, & \text{otherwise.} \end{cases}$$

Causal Logical Clock t

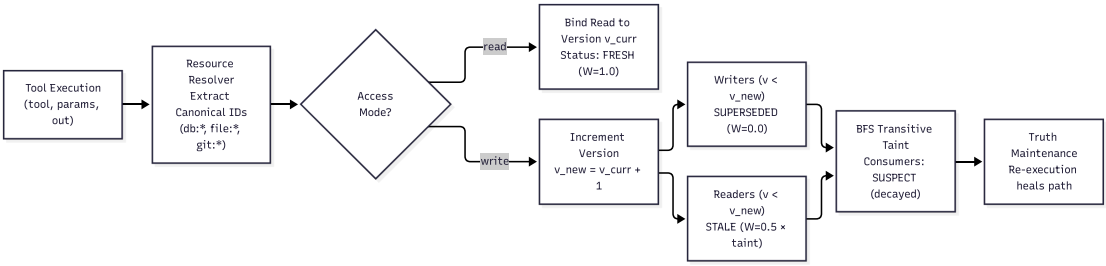
- Discrete causality steps, not noisy wall-clock time.
- Explicit version tracking across resources.
- Hard zeroing prevents obsolete state leakage.

Online Adaptive Decay Calibration

$$\lambda_{\text{task}} \leftarrow (1 - \eta)\lambda_{\text{task}} + \eta \frac{\ln 2}{\text{median}(\Delta t_{\text{useful}})}$$

- Learns optimal half-life per task from feedback.
- Rapid debugging $\rightarrow$ large λ; refactoring $\rightarrow$ small λ.

Contribution 2: Causal State Invalidation & Provenance Tracking



Direct Supersession ($W = 0$)

Newer write v_{new} supersedes older writers of the same resource immediately ($W = 0.0$).

Reader Staleness ($W = 0.5\tau$)

Earlier readers are marked `stale` when the resource they observed mutates ($W = 0.5 \times \tau$).

Transitive BFS Propagation

Consumers of invalid nodes receive attenuated taint and soft `suspect` tags.

Contribution 2 (Cont.): Hierarchical Macro-Action Mining

The Macro Distillation Pipeline

1. **Canonical Signatures:**

$$\sigma(a) = \langle \text{tool}, \text{shape}(\text{params}), \text{status} \rangle$$

   Strips literal values, retains structural intent.
2. **Sliding Window Clustering:** Slices executions into $W = 10$ windows; clusters matching sequences via Levenshtein matching.
3. **Slot Parameterization:** Extracts varying parameters into declarative variables: `{file_path}`, `{table}`.
4. **Promotion:** Mined skills with support ≥ 3 and success $\geq 80\%$ promoted to composite skills.

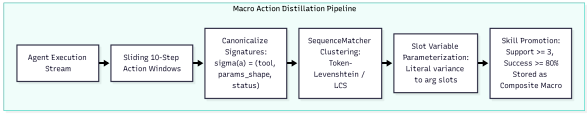
Macro Execution Benefits

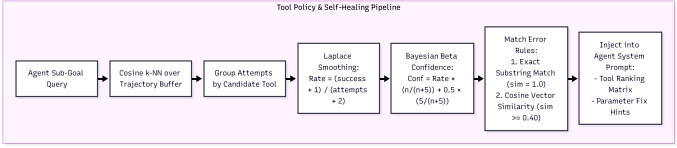
- **100% Tool-Sequence Match** on held-out test sessions.
- **100% Slot-Binding Accuracy** without LLM hallucinatory deviation.
- **53.2% Procedural Token Reduction** by substituting multi-step loops with a single declarative execution.

Hierarchical Skill Stacking

Level-2 atomic composites $\rightarrow$ chained into Level-3 multi-phase workflows (e.g., git branch $\rightarrow$ edit $\rightarrow$ test $\rightarrow$ commit).

Contribution 3: Policy Recommendation Matrix & Self-Healing





Laplace-Smoothed Tool Ranking

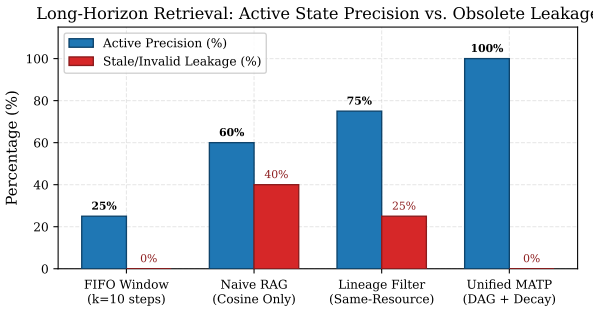
$$\hat{p}_T = \frac{k_{\text{success}} + 1}{N_{\text{attempts}} + 2}, \qquad c_T = \hat{p}_T \frac{N}{N+5} + 0.5\frac{5}{N+5}$$

Uniform Beta prior avoids 1/1 overconfidence. Bayesian shrinkage weights tools with empirical support. Color tiers: Green ($\geq 75\%$), Amber ($\geq 45\%$), Red ($< 45\%$).

Two-Tier Parameter Auto-Correction

- **Tier 1:** Exact error pattern substring match (sim $= 1.0$).
- **Tier 2:** Fuzzy vector cosine match (threshold $= 0.40$).
- Auto-injected into prompt *before* the tool is executed.

Experimental Evaluation: Eliminating Stale-State Hallucinations



40-Step Software Evolution Scenario

- 3 relational database schema migrations.
- 2 auth backend flips (basic $\rightarrow$ oauth2).
- Dependent ORM models, routes, test suites.

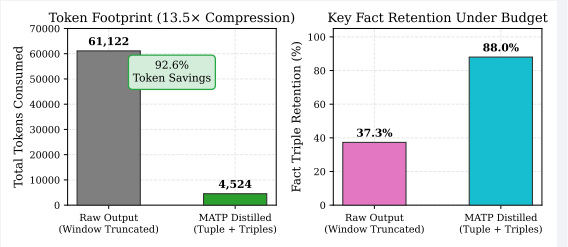
Strategy	Stale Leak	Active Prec
FIFO ($k = 10$)	0%	25.0%
Naive RAG	40%	60.0%
Lineage Only	25%	75.0%
Unified MATP	**0%**	**100.0%**

Key Finding

MATP completely eliminates obsolete schema hallucinations while retaining 100% active context.

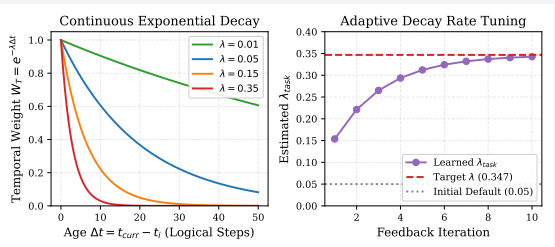
Distillation Efficiency & Adaptive Temporal Calibration

Token Reduction vs. Fact Retention



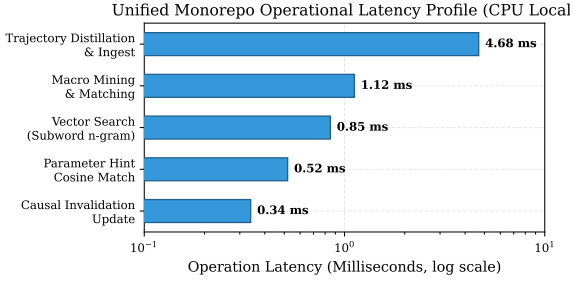
92.6% token reduction (13.5× compression) while preserving **88.0%** essential fact triples (vs. 37.3% for raw text).

Temporal Decay & Adaptive Tuning



Smooth exponential decay $e^{-\lambda \Delta t}$. Online tuner converges to optimal task half-life within 5 feedback iterations.

Operational Latency Profile: Sub-5 ms Local Execution



Zero-GPU Commodity CPU Benchmark

- **Causal DAG Invalidation:** 0.34 ms
- **Parameter Hint Search:** 0.52 ms
- **Subword Vector Retrieval:** 0.85 ms
- **Macro Skill Matching:** 1.12 ms
- **Hot-Path Ingestion:** 4.68 ms

Monorepo Verification Status

- **Single-file durability:** ACID SQLite schema.
- **Zero external services:** Deterministic local subword vectorizer.
- **Live demo verified:** Full round-trip execution passing with zero errors.

Summary of Deliverables & Conclusion

Consolidated Deliverables

- **Unified Monorepo:** Merged Python & TypeScript MCP servers into `matp-monorepo`.
- **Durable SQLite Engine:** 8 relational tables with causal DAG, macro skills, and decay tuning.
- **Automated Tools:** 20 MCP endpoints spanning memory, lineage, macros, and policy.
- **Publication Report:** Full IEEE Computer Society 8-page paper rendered with Tectonic.

Core Technical Impact

- **100% Active Precision:** Zero stale vector leakage.
- **13.5× Compression:** 92.6% token savings with 88.0% factual fidelity.
- **Self-Healing Policy:** Beta confidence ranking and auto-injected schema hints.
- **Sub-5 ms Latency:** Instantaneous local stdio responses.

Thank You! Questions & Discussion.